

OBJECT-ORIENTED PROGRAMMING

LAB GUIDELINES

Semester 20261

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1. Lab Guidelines

1. Students must follow the instructions provided in the lab materials and complete all exercises.

If students are unable to complete the exercises during the scheduled lab session, they must continue working on them at home.

2. For each lab, students must submit their work twice:

- a. **Immediately after the lab session:**

At this time, students must upload to GitHub all work completed during the lab session.

- b. **10:00 PM on the day before the next class:**

At this time, students must complete all tasks and exercises in the lab materials, including guided exercises and exercises that already provide sample source code.

Students must commit and push all work to the **master** or **main** branch of a valid repository as described in Section 3.

Each student must complete and submit their own work.

If a lab contains questions, students must write answers to all questions in a text file named:

answers.txt

and submit this file in their repository.

2. Git and GitHub Guidelines

2.1. Git Learning Resources

Students can learn Git commands from the following resources:

- i. <https://git-scm.com/book/en/v2>
- ii. <https://git-scm.com/docs/gittutorial>
- iii. <https://www.atlassian.com/git/tutorials>
- iv. <https://www.youtube.com/watch?v=HVsySz-h9r4>

2.2. GitHub Learning Resources

- i. <https://www.freecodecamp.org/news/git-and-github-for-beginners/>
- ii. <https://www.youtube.com/watch?v=RG0j5yH7evk>

2.3. GitHub Authentication

Students can authenticate with GitHub using a Personal Access Token (PAT) or SSH.

- i. Create a Personal Access Token:

<https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/creating-a-personal-access-token>

- ii. Create an SSH key:

<https://docs.github.com/en/authentication/connecting-to-github-with-ssh>

3. Using Git/GitHub for Submission

3.1. Installing Git

Install Git on your computer.

Students may also use optional tools such as SourceTree or eGit for Eclipse.

Git tutorial:

<https://www.youtube.com/watch?v=HVsySz-h9r4>

3.2. Creating a GitHub Account

Create an account at:

<https://github.com/>

3.3. Creating a Repository

Create a **public repository** using the following naming convention:

`OOP.Lab.20261.StudentID.StudentName`

For example:

```
OOP.Lab.20261.20162912.NguyenHoangTu
```

If this naming convention is not followed, the student's repository may be skipped during the submission review process.

3.4. Creating a Personal Access Token or Configuring SSH

Create a Personal Access Token (PAT) or configure SSH authentication.

Students can follow the instructions in Section 5 to create and configure a PAT for GitHub.

3.5. Cloning the Repository

Students can clone the repository and work on the default branch (**master/main**) or create another branch.

Navigate to the working directory on your computer:

```
cd <local_working_folder>
```

Clone the repository:

```
git clone <repository_url>
```

Navigate to the repository:

```
cd <repository_name>
```

For example:

```
cd OOP.Lab.20261.20162912.NguyenHoangTu
```

Check the current branch:

```
git branch
```

Create and switch to a new branch:

```
git checkout -b <your_branch>
```

Students can then make changes such as creating new files, editing existing files, deleting files, or making other modifications.

3.6. Committing and Pushing Changes

Navigate to the repository on your computer:

```
cd <local_working_folder/repository_name>
```

Stage all changes:

```
git add -A
```

The `-A` option stages all changes, including newly created, modified, and deleted files.

Students can also use:

```
git add .
```

Check the repository status:

```
git status
```

Commit the changes:

```
git commit -m "Your comment for the update"
```

Push commits from the local repository to the remote repository:

```
git push origin <your_branch>
```

4. Pulling Changes

4.1. Pulling Without Conflicts

If you want to temporarily save your current changes:

```
git stash
```

Then pull from the remote repository:

```
git pull origin <your_branch>
```

4.2. Pulling and Resolving Conflicts

After committing and pushing changes to the remote repository, conflicts may occur when branches contain different changes to the same resource.

For example:

```
git checkout master
git pull origin master
git checkout <your_branch>
git rebase master
git add -A
git status
git rebase --continue
```

When a conflict occurs, Git may display both versions in the conflicted file.

For example:

```
<<<<<< HEAD
suggestor.setClientId(clientID);
suggestion =
suggestor.translate(input).replaceAll("-", " ");
=====

suggestion = suggestor.translate(input);

> > > > > Your comment for the update
> > > > >
```

Students must inspect the conflicting versions and decide which version should be kept.

For example, after resolving the conflict:

```
suggestion =
suggestor.translate(input).replaceAll("-", " ");
```

After resolving the conflict, continue working and push the updated version to the remote repository:

```
git push origin <your_branch> -f
```

5. Creating a Personal Access Token on GitHub

From your GitHub account, navigate to:

Settings → Developer Settings → Personal Access Token → Generate New Token

Enter your password if requested, provide the required information, generate the token, and copy the generated token.

The token may look similar to:

```
ghp_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

Students must keep their Personal Access Token secure.

5.1. Windows Operating System

Open:

Control Panel → Credential Manager → Windows Credentials

Find:

```
git:https://github.com
```

Select **Edit** and replace the password with your GitHub Personal Access Token.

If you cannot find the GitHub credential, select:

Add a generic credential

Use:

```
Internet address:  
git:https://github.com
```

Then enter your GitHub username and use the Personal Access Token as the password.

Select **OK** to save the credentials.

5.2. macOS Operating System

Click the Spotlight icon and search for:

```
Keychain Access
```

Open Keychain Access and search for:

```
github.com
```

Find the corresponding Internet Password entry and edit or delete the credentials if necessary.

5.3. Linux Operating System

Configure the Git client on your computer with your username and email address:

```
git config --global user.name "your_github_username"  
git config --global user.email "your_github_email"  
git config -l
```

After Git has been configured, students can clone a GitHub repository.

For example:

```
git clone https://github.com/YOUR-USERNAME/YOUR-REPOSITORY
```

The command may request authentication information:

```
Username: <type your username>  
Password: <type your password or personal access token>
```

To cache authentication credentials:

```
git config --global credential.helper cache
```

If necessary, the cached credential configuration can be removed:

```
git config --global --unset credential.helper  
git config --system --unset credential.helper
```

Check the configuration using:

```
git pull -v
```

6. Using a Personal Access Token in PhpStorm

If you are using PhpStorm, open:

Git → Pull

and select authentication using a Personal Access Token.

Enter the PAT to authenticate and perform pull/push operations with Git.